

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Advanced)

UNIT TEST

29-11-2020

JEE(Main + Advanced) : NURTURE COURSE [PHASE : S, I(A), I, II, III, IV(A) & IV]**ANSWER KEY****PAPER-1****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6
	A	A,B	A,C	B,C,D	A,C	A,B,C	A,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A	3.25	0.73	6.00	7.50	2.40	1.25
SECTION-III	Q.	1	2	3	4	5	6
	A	3	9	4	7	4	5

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6
	A	B,C	A,B,C,D	A,B,C,D	A,B,D	C	A,B,D
SECTION-II	Q.	1	2	3	4	5	6
	A	24.00	64.00	4.00	18.00	5.00	6.00
SECTION-III	Q.	1	2	3	4	5	6
	A	1	4	6	2	3	6

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6
	A	A,B,D	B,D	C,D	A,B	A,C	A,B,D
SECTION-II	Q.	1	2	3	4	5	6
	A	3.00	0.00	312.50	194.00	16.66 or 16.67	3.70 or 3.71
SECTION-III	Q.	1	2	3	4	5	6
	A	6	1	7	9	4	6

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JEE(Main)

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JEE(Main + Advanced) : NURTURE COURSE [PHASE : S, I(A), I, II, III, IV(A) & IV]**ANSWER KEY****PAPER-2****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	D	A	D	D	B	C	C	A	C
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	D	B	D	B	B	A	A	A	D
SECTION-II	Q.	1	2	3	4	5					
	A.	2.00	6.00	2.00	18.00	5.00					

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	C	A	D	C	B	A	C	C	C
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	D	C	B	C	C	D	A	D	D
SECTION-II	Q.	1	2	3	4	5					
	A.	1.25	9.00	9.00	12.00	2.00					

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	D	D	A	C	A	A	C	B	A
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	A	C	C	B	C	B	B	D	D
SECTION-II	Q.	1	2	3	4	5					
	A.	1.00	3.00	4.00	24.00	0.00					

JEE(Main + Advanced) : NURTURE COURSE [PHASE : S, I(A), I, II, III, IV(A) & IV]**PAPER-1****PART-1 : PHYSICS****SOLUTION****SECTION-I**1. **Ans. (A,B)**

$$F - f = ma$$

$$mg - \frac{mg}{2} = ma \Rightarrow a = \frac{g}{2}$$

At mid point



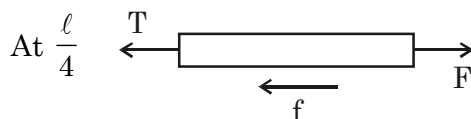
$$T - f = \frac{m}{2}a$$

$$T - \frac{mg}{4} = \frac{mg}{4} \quad T = \frac{mg}{2}$$

$$\text{Stress} = \frac{mg}{2A} \quad \text{Strain} = \frac{mg}{2Ay}$$

$$\text{Energy density} = \frac{1}{2} \text{Stress} \times \text{strain}$$

$$= \frac{m^2 g^2}{8A^2 y}$$



$$F - T - \frac{mg}{8} = \frac{m}{4}a$$

$$mg - T - \frac{mg}{8} = \frac{mg}{8}$$

$$T = \frac{3mg}{4}$$

$$\text{Stress} = \frac{3mg}{4A}$$

2. **Ans. (A,C)**

Sol. Let escalator moving with speed v_0 and boy moving with speed v w.r.t. ground

$$v + v_0 = \frac{3}{2}v_0$$

$$v = \frac{v_0}{2} = 3 \text{ m/s}$$

Speed of boy is half the speed escalator when boy goes up by h , escalator move down by $2h$.

Work done by boy on escalator

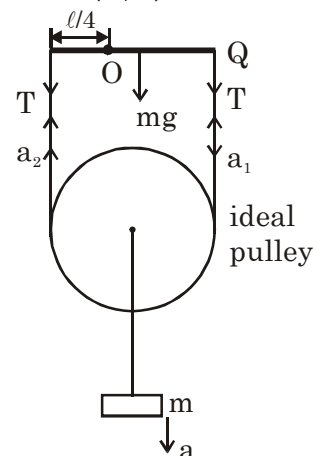
$$= N \cdot 2h = 2mgh$$

$$= 2 \times 40 \times 10 \times 12 = 9600 \text{ J}$$

$$\text{Total work by boy} = 3mgh = 3 \times 40 \times 10 \times 12 = 14400 \text{ J} = 14.4 \text{ kJ}$$

3. **Ans. (B,C,D)**

Sol. On giving heat both sphere expand, sphere A expand in upward direction and B in downward, hence potential energy of A increases while pot energy of B decreases hence heat given to A not only increases temp but also PE, so temperature of A increase less than B.

4. **Ans. (A,C)**

Sol.

For rod

$$\tau_0 = I_0 \alpha$$

$$mg \frac{\ell}{4} + T \frac{3\ell}{4} - T \frac{\ell}{4} = \left(\frac{m\ell^2}{12} + \frac{m\ell^2}{16} \right) \alpha$$

$$mg \frac{\ell}{4} + \frac{T\ell}{2} = \frac{7m\ell^2}{48} \alpha$$

$$mg + 2T = \frac{7m\ell}{12} \alpha \quad \dots (i)$$

$$a_1 = \frac{3\ell}{4} \alpha, a_2 = \frac{\ell}{4} \alpha$$

$$a = \frac{a_1 - a_2}{2} = \frac{\ell \alpha}{4}$$

for block $mg - 2T = ma$

$$mg + 2T = \frac{7ma}{3} \text{ from equation (i)}$$

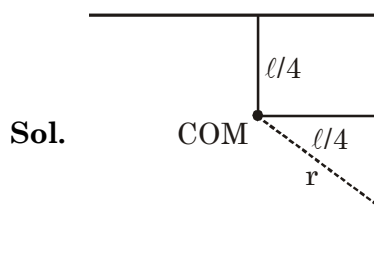
$$2mg = \frac{10ma}{3}$$

$$a = \frac{3g}{5}$$

$$a = \frac{\ell \alpha}{4} \Rightarrow \frac{4\alpha}{4} = \frac{3 \times 10}{5}$$

$$\alpha = 6 \text{ rad/sec}^2$$

5. **Ans. (A,B,C)**



$$r = \sqrt{\frac{\ell^2}{16} + \frac{\ell^2}{16}} = \frac{\ell}{4} \sqrt{2} = \frac{\ell}{2\sqrt{2}}$$

By angular momentum conservation about COM of system

$$\frac{mv\ell}{4} = I_C \omega$$

$$\frac{mv\ell}{4} = \frac{5m\ell^2}{12} \omega$$

$$\omega = \frac{3v}{5\ell}$$

$$I_C = 2 \left(\frac{m\ell^2}{12} + \frac{m\ell^2}{8} \right)$$

$$= \frac{5m\ell^2}{12}$$

By momentum conservation $mv = 2mv_C$

$$v_C = \frac{v}{2}$$

distance of IAOR from COM

$$x = \frac{v_C}{\omega} = \frac{v\ell}{2 \times 3v} = \frac{5\ell}{6}$$

Loss in KE = $k_i - k_f$

$$= \frac{1}{2} mv^2 - \left(\frac{1}{2} (2mv_C^2) + \frac{1}{2} I_C \omega^2 \right)$$

$$= \frac{7}{40} mv^2$$

6. **Ans. (A,C,D)**

Sol. Ball is striking at mid point of each step, let horizontal velocity of ball is v_x . Since each time ball rises to same height, hence ball strike & rebound from each step with same vertical speed. Let vertical velocity just before collision is v_y just after collision

$$v_y^1 = ev_y = \frac{2}{3} v_y$$

Time between two collision

$$T = \frac{\ell}{v_x} \text{ by horizontal motion}$$

$$v_y^1 = -v_y + a_y t \Rightarrow t = \frac{v_y^1 + v_y}{g}$$

$$\frac{\ell}{v_x} = \frac{v_y(1+e)}{g} \quad \dots (i)$$

By applying w.e.t. between two collisions

$$mg\ell = \frac{1}{2} m(v_x^2 + v_y^2) - \frac{1}{2} m(v_x^2 + v_y'^2)$$

$$v_y^2 = v_y'^2 + 2g\ell$$

$$v_y^2(1 - e^2) = 2g\ell$$

$$v_y = \sqrt{\frac{2g\ell}{1 - e^2}} = \sqrt{\frac{2 \times 10 \times 4}{1 - \frac{4}{9}}}$$

$$v_y = \sqrt{2 \times 2 \times 4 \times 9} = 4 \times 3 = 12 \text{ m/s}$$

$$v_y' = ev_y = 2/3 \times 12 = 8 \text{ m/s}$$

from eq. (i)

$$\frac{4}{v_x} = \frac{12 \left(1 + \frac{2}{3}\right)}{10}$$

$$v_x = \frac{10 \times 4 \times 3}{12 \times 5} = 2 \text{ m/s}$$

Time between two collisions

$$= \frac{\ell}{v_x} = \frac{4}{2} = 2 \text{ sec}$$

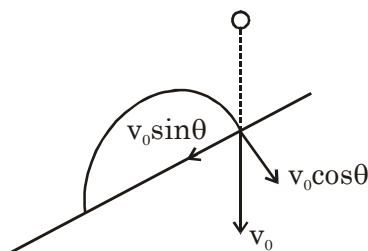
Maximum height above each step

$$h = \frac{v_y'^2}{2g} = \frac{8 \times 8}{2 \times 10} = 3.2 \text{ m}$$

SECTION-II

1. Ans. 3.25

Sol.



Since collision is elastic ball rebound with same speed, after collision motion is projectile on incline plane

Let t - time of flight

$$t = \frac{2u_y}{a_y} = \frac{2v_0 \cos \theta}{g \cos \theta}$$

$$= \frac{2v_0}{g}$$

$$R = v_0 \sin \theta t + \frac{1}{2} g \sin \theta t^2$$

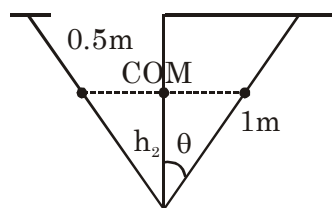
As velocity along the plane is increasing linearly so R_1, R_2, R_3 are in A.M.

$$R_2 = \frac{R_1 + R_3}{2} = \frac{a(R_1 + R_3)}{6.5}$$

$$a = 3.25$$

2. Ans. 0.73

Sol.



$$\text{Work done} = mg \Delta h_{\text{cm}}$$

$$0.4 = 2 \times 10 \Delta h_{\text{cm}}$$

$$\Delta h = .02 \text{ m} = 2 \text{ cm (upward)}$$

COM move 2 cm up when mid point is pulled down.

In figure 2

$$\cos \theta = \frac{0.5}{1} = \frac{h_2}{0.5}$$

$$h_2 = .25 = 25 \text{ cm}$$

$$\Delta h = h_2 - h_1$$

$$2 = 25 - h_1$$

$$h_1 = 23 \text{ cm}$$

Total height of COM from ground before pulling

$$h_{\text{cm}} = 50 + 23 = 73 \text{ cm}$$

$$= 0.73 \text{ m}$$

3. Ans. 6.00

Sol. Let m mass of ice is melted

$$\Delta Q_{\text{ice}} = \Delta Q_{\text{water}}$$

$$12 \times 0.5 \times 20 + m \times 80 = 30 \times 1 \times 20$$

$$120 + 80 m = 600$$

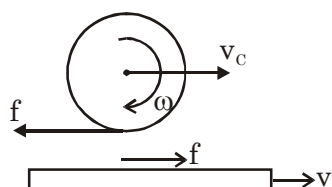
$$80 m = 480$$

$$m = 6 \text{ gm}$$

$$\text{mass remaining} = 12 - 6 = 6 \text{ gm}$$

4. Ans. 7.50

Sol.



$$\text{Friction of cylinder } f = \mu mg$$

$$\text{For cylinder } v_C = v_0 - \mu gt$$

$$\text{For plank } v_P = \mu gt$$

$$\text{For rolling } v_C - r\omega = v_P$$

$$\text{For rolling } v_C - r\omega = v_P \dots (i)$$

$$\tau = I\alpha$$

$$\mu m g r = \frac{m r^2}{2} \alpha$$

$$\alpha = \frac{2 m g}{r}$$

$$\omega = \alpha t = \frac{2 \mu g t}{r}$$

By eq. (i)

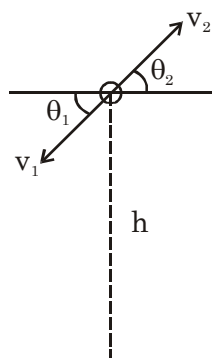
$$v_0 - \mu g t - 2 \mu g t = \mu g t$$

$$t = \frac{v_0}{4 \mu g}$$

$$v_0 = v_0 - \mu g t = v_0 - \frac{v_0}{4} = \frac{3 v_0}{4}$$

$$v_c = \frac{3 \times 10}{4} = 7.50 \text{ m/s}$$

5. Ans. 2.40



Sol.

Bomb explode at highest point

$$\left. \begin{array}{l} p_i = 0 \\ p_f = 0 \end{array} \right\} \text{ just before and after explosion.}$$

Since both travel same horizontal distance

$$v_1 \cos \theta_1 t_1 = v_2 \cos \theta_2 t_2$$

$$\frac{v_1 \cos \theta_1}{v_2 \cos \theta_2} = \frac{t_2}{t_1}$$

By momentum conservation

$$m_1 v_1 \cos \theta_1 = m_2 v_2 \cos \theta_2$$

$$\frac{v_1 \cos \theta_1}{v_2 \cos \theta_2} = \frac{m_2}{m_1} = 2 = \frac{t_2}{t_1}$$

$$t_2 = 2 t_1 \text{ \& } t_2 - t_1 = 4$$

$$t_1 = 4 \quad t_2 = 8$$

$$\frac{t_1 + t_2}{5} = \frac{12}{5} = 2.40$$

6. Ans. 1.25

Sol. Rate of energy loss by block body

$$E_1 = \sigma 4 \pi r^2 (T^4 - T_0^4)$$

When block body is enclosed by spherical shell, let temp of shell is T_1

$$E_2 = \sigma 4 \pi r^2 (T^4 - T_1^4)$$

Also

$$\sigma 4 \pi r^2 (T^4 - T_1^4) = \sigma 4 \pi R^2 (T_1^4 - T_0^4)$$

$$T_1^4 = \frac{r^2 T^4 + R^2 T_0^4}{R^2 + r^2}$$

$$E_2 = \sigma 4 \pi r^2 \left[T^4 - \frac{r^2 T^4 + R^2 T_0^4}{R^2 + r^2} \right]$$

$$E_2 = \frac{\sigma 4 \pi r^2}{R^2 + r^2} \left[R^2 (T^4 - T_0^4) \right]$$

$$\frac{E_1}{E_2} = \frac{R^2 + r^2}{R^2} = \frac{4 + 1}{4} = \frac{5}{4}$$

$$= 1.25$$

SECTION-III

1. Ans. 3

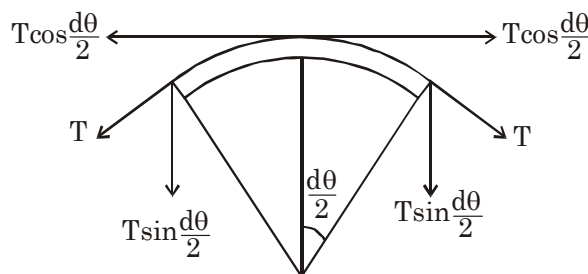
Sol. Speed of temperature graph = temperature gradient

$$\text{Temperature gradient} = -\frac{i}{kA} = \tan \theta$$

$$\frac{\tan \theta_1}{\tan \theta_2} = \frac{k_c}{k_1} = \frac{\sqrt{3}}{\frac{1}{\sqrt{3}}} = 3$$

2. Ans. 9

Sol.



$$2 T \sin \frac{d\theta}{2} = dm R \omega^2 = \rho A R d\theta \omega^2$$

$$T = \rho A R^2 \omega^2 \Rightarrow \frac{T}{A} = \rho R^2 \omega^2$$

By Hook's law $\frac{T}{A} = \frac{Y\Delta\ell}{\ell}$

$$\Delta\ell = \frac{\rho R^2 \omega^2 \times 2\pi R}{E}$$

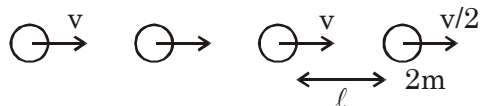
$$\Delta\ell = \frac{2\pi R^3 \rho \omega^2}{E} = \frac{18\pi R^3 \rho \omega^2}{aE}$$

$$a = 9$$

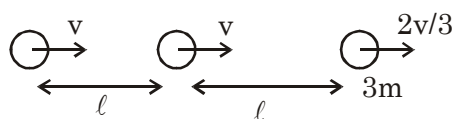
3. Ans. 4



time in first collision $t_1 = \frac{\ell}{v}$



Time in 2nd collision $t_2 = \frac{\ell}{v/2} = \frac{2\ell}{v}$



Time in 3rd collision $= \frac{\ell}{\frac{v}{3}} = \frac{3\ell}{v}$,

similarly for 4th collision time $= \frac{4\ell}{v}$

Total time $= \frac{\ell}{v} + \frac{2\ell}{v} + \frac{3\ell}{v} + \frac{4\ell}{v} = \frac{10\ell}{v}$

$$= \frac{10\ell}{v} = \frac{2.5a\ell}{v} \Rightarrow a = 4$$

4. Ans. 7

Sol. For rolling motion $f = \frac{mg \sin \theta}{1 + \frac{mR^2}{I_c}}$

$$f = \frac{mg \sin \theta}{1 + 2} = \frac{1}{3} mg \sin \theta$$

$$f \leq f_{\max}$$

$$\frac{1}{3} mg \sin \theta \leq \mu mg \cos \theta$$

$$\tan \theta \leq 3\mu$$

$$\tan \theta_{\max} = 3\mu$$

$$\theta_{\max} = \tan^{-1} \left(3 \times \frac{2}{15} \right)$$

$$\theta_{\max} = \tan^{-1} \left(\frac{2}{5} \right)$$

$$\alpha + \beta = 2 + 5 = 7$$

5. Ans. 4

Sol. By conservation of angular momentum about point P

Let speed of bug w.r.t. ground is v_0

$$L_i = L_f = 0 \quad m_{\text{ring}} = 2m_{\text{bug}}$$

$$L_{\text{ring}} = L_{\text{bug}} \quad m_{\text{bug}} = m$$

$$4mR^2\omega = m2R v_0$$

$$\omega = \frac{v_0}{2R}$$

By relative motion $v_0 + 2R\omega = v$

$$v_0 + v_0 = v$$

$$v_0 = \frac{v}{2} = \frac{8}{2} = 4 \text{ m/s}$$

6. Ans. 5

Sol. In figure (a)

$$I_1 = \frac{mR^2}{2}$$

In figure (b)

$$I_2 = \frac{mR^2}{4} + \left(\frac{mR^2}{4} + \frac{m}{2} (2R)^2 \right)$$

$$= \frac{mR^2}{2} + 2mR^2 = \frac{5mR^2}{2}$$

$$\frac{I_2}{I_1} = 5$$

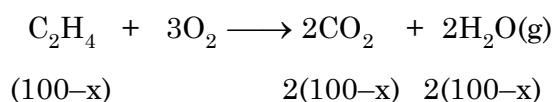
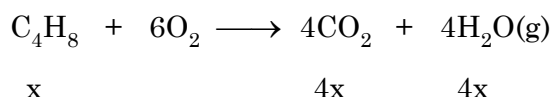
PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans.(B,C)

2. Ans.(A,B,C,D)



So for any value of x $\{0 < x < 100\}$

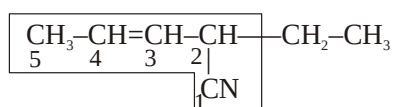
always, $V_{\text{CO}_2} = V_{\text{H}_2\text{O}}$

3. Ans (A,B,C,D)

4. Ans.(A,B,D)

5. Ans.(C)

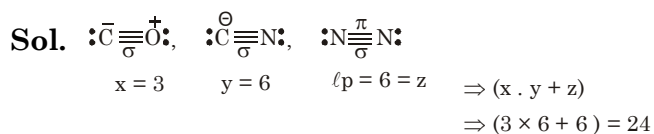
6. Ans. (A,B,D)



2-Ethyl pent-3-ene nitrile

SECTION-II

1. Ans.(24.00)



2. Ans. (64.00)

3. Ans (4.00)

4. Ans.(18.00)

5. Ans.(5.00)

Visible lines belongs to Balmer series

$$\begin{aligned} 0.50, \text{ no. of line} &= n - 2 \\ &= 7 - 2 = 5 \end{aligned}$$

6. Ans (6.00)

SECTION-III

1. Ans.(1)

2. Ans.(4)

Sol. $2.22 \text{ mg CaCl}_2 \equiv 2 \text{ mg CaCO}_3$

$1.9 \text{ mg MgCl}_2 \equiv 2 \text{ mg CaCO}_3$

Total $\text{CaCO}_3 = 4 \text{ mg ; ppm}$

$$= \frac{4 \times 10^{-3}}{10^3} \times 10^6 = 4 \text{ ppm}$$

3. Ans.(6)

4. Ans.(2)

5. Ans. (3)

I.E. of H-atom = 13.6 eV

= photon energy in bulb A

I.E. of He^+ ions

= $(13.6 \times 2^2) \text{ eV}$

= photon energy in bulb B

In bulb A power

$$= 30 = n_1 \times (13.6 \times 1.6 \times 10^{-19})$$

In bulb B power

$$= 40 = n_2 \times (13.6 \times 4 \times 1.6 \times 10^{-19})$$

$$\frac{30}{40} = \frac{n_1}{4 \times n_2}$$

$$n_1 : n_2 = 3 : 1$$

6. Ans.(6)

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (A,B,D)**

Sol. Let $x = \cos \theta$

$$4 \cos^3 \theta - 3 \cos \theta = -\frac{\sqrt{3}}{2}$$

$$\cos 3\theta = \cos \frac{5\pi}{6} \Rightarrow 3\theta = 2n\pi \pm \frac{5\pi}{6}$$

$$\Rightarrow \theta = \frac{2n\pi}{3} \pm \frac{5\pi}{18}$$

$$\text{put } n = 0, \theta = \frac{5\pi}{18}$$

$$n = 1, \theta = \frac{2\pi}{3} + \frac{5\pi}{18} = \frac{17\pi}{18}; \theta$$

$$= \frac{2\pi}{3} - \frac{5\pi}{18} = \frac{7\pi}{18}$$

2. **Ans. (B,D)**

Sol. $a - d, a, a + d \Rightarrow a = 3$

$$(a - d)^2 + a^2 + (a + d)^2 = 35$$

$$\Rightarrow 3a^2 + 2d^2 = 35 \Rightarrow 2d^2 = 8 \Rightarrow d = \pm 2$$

$$\text{for } d = 2; S_n = n^2; a = 1$$

$$\text{for } d = -2; S_n = -n^2 + 6n; a = 5$$

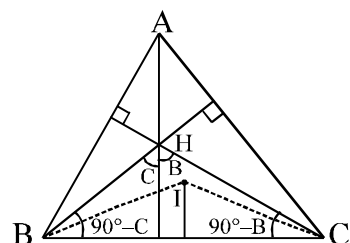
3. **Ans. (C,D)**

Sol. In any $\triangle ABC$, angle subtended by the side BC at orthocentre and incentre are

$$B + C \text{ and } 180^\circ - \left(\frac{B+C}{2}\right) \text{ respectively.}$$

$$\text{Now, } B + C = 180^\circ - \left(\frac{B+C}{2}\right)$$

(Given)



$$\Rightarrow B + C = 120^\circ \Rightarrow A = 60^\circ$$

$$\text{Similarly, } B = C = 60^\circ$$

$$\Rightarrow \triangle ABC \text{ is equilateral.}$$

4. **Ans. (A,B)**

Sol. Let D_1, D_2 and D_3 be the discriminants of $P(x), Q(x)$ and $R(x)$ respectively, then

$$D_1 = b^2 - 4ac$$

$$D_2 = c^2 - 4ab$$

$$\text{and } D_3 = (bc)^2 - 4a(b^3 + c^3 - 4abc)$$

$$= b^2c^2 - 4ab^3 - 4ac^3 + 16a^2bc$$

$$= b^2(c^2 - 4ab) - 4ac(c^2 - 4ab)$$

$$= (b^2 - 4ac)(c^2 - 4ab) = D_1 \cdot D_2$$

Since $R(x) = 0$ has non-real roots.

$$\therefore D_3 < 0 \Rightarrow D_1 \cdot D_2 < 0$$

$$\Rightarrow \text{Either } D_1 < 0 \text{ and } D_2 > 0 \text{ or}$$

$$D_1 > 0 \text{ and } D_2 < 0$$

Hence either $P(x) = 0$ has distinct real roots and $Q(x) = 0$ has non-real roots.

OR

$P(x) = 0$ has non-real roots and $Q(x) = 0$ has distinct real roots.

5. **Ans. (A,C)**

Sol. Let $x + 5 = 14 \cos \theta$

$$y - 12 = 14 \sin \theta$$

$$\therefore x^2 + y^2 = (14 \cos \theta - 5)^2 + (14 \sin \theta + 12)^2$$

$$= 196 + 25 + 144 + 28(12 \sin \theta - 5 \cos \theta)$$

$$= 365 + 28(12 \sin \theta - 5 \cos \theta)$$

$$\therefore \sqrt{x^2 + y^2} \Big|_{\min} = \sqrt{365 - 28 \times 13}$$

$$= \sqrt{365 - 364} = 1$$

Ans. $\Rightarrow A, C$

6. **Ans. (A,B,D)**

Sol.

(A) For $b = 9$ solving line and parabola

$$2x - 9 = x^2 - 4x$$

$$\Rightarrow x^2 - 6x + 9 = 0$$

$$\Rightarrow (x - 3)^2 = 0$$

$\therefore$ line is tangent to parabola $\Rightarrow$ (A) is correct

(B) For $b = 7$, line is $y = 2x - 7$

Homogenising parabola with line

$$x^2 - 4x \left(\frac{2x - y}{7} \right) - y \left(\frac{2x - y}{7} \right) = 0$$

coefficient of x^2 + coefficient of $y^2 = 0$

$$\Rightarrow 1 - \frac{8}{7} + \frac{1}{7} = 0$$

$\therefore \angle AOB$ is $90^\circ \Rightarrow$ (B) is correct

(C) Homogenising the parabola with line, we get

$$x^2 - 4x \left(\frac{2x-y}{b} \right) - y \left(\frac{2x-y}{b} \right) = 0$$

for x-axis to bisect $\angle COD$
coefficient of xy should be zero

$$\text{i.e. } \frac{4}{b} - \frac{2}{b} = 0$$

which is not possible for any $b \in \mathbb{R} \Rightarrow$ (C) is false

(D) For $b > 9$

Solving line and parabola

$$x^2 - 6x + b = 0$$

$$D = 36 - 4b$$

If $b > 9$

then $D < 0$

$\therefore$ line passes outside the parabola $\Rightarrow$ (D) is correct.

SECTION-II

1. Ans. 3.00

Sol. Let (h, k) be point of intersection of tangents then chord of contact is

$$yk = 4(x + h)$$

$$4x - yk + 4h = 0 \quad \dots (i)$$

But given line is

$$x - y - 1 = 0 \quad \dots (ii)$$

Comparing (i) and (ii)

$$\therefore \frac{4}{1} = \frac{-k}{-1} = \frac{4h}{-1} \Rightarrow h = -1, k = 4$$

$\therefore$ point $\equiv (-1, 4)$

2. Ans. 0.00

Sol. $\therefore$ Equation of ellipse is $9x^2 + 16y^2 = 144$

$$\text{or } \frac{x^2}{16} + \frac{y^2}{9} = 1$$

Comparing this with $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ then we

get $a^2 = 16$ and $b^2 = 9$

and comparing the line $y = x + \lambda$ with

$$y = mx + c$$

$\therefore m = 1$ and $c = \lambda$

If the line $y = x + \lambda$ touches the ellipse

$$9x^2 + 16y^2 = 144, \text{ then } c^2 = a^2m^2 + b^2$$

$$\Rightarrow \lambda^2 = 16 \times 1^2 + 9 \Rightarrow \lambda^2 = 25$$

$\therefore \lambda = \pm 5$ Ans.

3. Ans. 312.50

Sol. $y = (7 \cos x + 24 \sin x)(7 \sin x - 24 \cos x)$

$$r \cos \phi = 7 ; r \sin \phi = 24$$

$$r^2 = 625 ; \tan \phi = \frac{24}{7}$$

$$y = r \cos(x - \phi) \cdot r \sin(\theta - x)$$

$$= \frac{r^2}{2} \cdot 2 \sin(x - \phi) \cos(x - \phi) = \frac{r^2}{2} \cdot (\sin 2(x - \phi))$$

$$y_{\max} = \frac{25^2}{2} = \frac{625}{2}$$

4. Ans. 194.00

Sol. Given $a_1 = 2 ; \frac{a_n}{a_{n-1}} = \frac{a_{n-1}}{a_{n-2}} \Rightarrow a_1, a_2, a_3, a_4, a_5, \dots$ in G.P.

Let $a_2 = x$ then for $n = 3$ we have

$$\frac{a_3}{a_2} = \frac{a_2}{a_1} \Rightarrow a_1^2 = a_2 a_3 \Rightarrow a_3 = \frac{x^2}{2}$$

i.e. $2, x, \frac{x^2}{2}, \frac{x^3}{4}, \frac{x^4}{8}, \dots$ with common

$$\text{ratio } r = \frac{x}{2}$$

$$\text{given } \frac{x^4}{8} \leq 162 \Rightarrow x^4 \leq 1296 \Rightarrow x \leq 6$$

Also x and $\frac{x^4}{8}$ are integers $\Rightarrow x$ must be

even then only $\frac{x^4}{8}$ will be an integer.

hence possible values of x is 4 and 6.

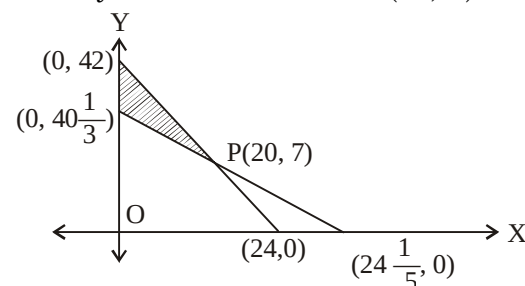
($x \neq 2$ as terms are distinct)

hence possible value of $a_5 = \frac{x^4}{8}$ is $\frac{4^4}{8}, \frac{6^4}{8}$

32, 162 $\Rightarrow$ B, D

5. Ans. 16.66 or 16.67

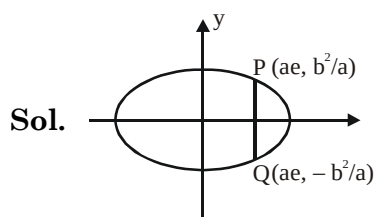
Sol. Clearly lines intersect at $P(20, 7)$



$$\therefore \text{Area of shaded region} = \frac{1}{2} \left(42 - 40 \frac{1}{3} \right) 20$$

$$= \frac{1}{2} \left(\frac{5}{3} \right) 20 = \frac{50}{3} \text{ (square units)}$$

6. **Ans. 3.70 or 3.71**



Equation of tangent at P is

$$ex + y = a$$

$$\Rightarrow e = \frac{2}{3}, a = \frac{10}{3}$$

$$\text{and } b = \frac{10\sqrt{5}}{9}$$

$$\text{length of latus rectum} = \frac{2b^2}{a} = \frac{100}{27}$$

SECTION-III

1. **Ans. 6**

Sol. $\tan^2 x = \frac{1}{3} \Rightarrow \tan x = \frac{1}{\sqrt{3}} \text{ or } -\frac{1}{\sqrt{3}}$

$$\Rightarrow \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6}$$

$\Rightarrow 6$ solutions **Ans.**

2. **Ans. 1**

Sol. Let V_1 and V_2 be the vertices of the parabola $f(x)$ and $g(x)$ respectively.

We have $V_1 \equiv (1, a)$

$$\text{and } V_2 \equiv (a, -a^2 + 10 - 2a)$$

$$\left[\begin{array}{l} \text{hence max. value of } f(x) = a \\ \text{and min. value of } g(x) = b - 2a - a^2 \end{array} \right]$$

$$\text{Now, } a < -a^2 + 10 - 2a$$

$$\Rightarrow a^2 + 3a - 10 < 0$$

$$\therefore (a + 5)(a - 2) < 0$$

$$\Rightarrow -5 < a < 2$$

$\therefore$ Largest natural number $a = 1$ **Ans.**

3. **Ans. 7**

Sol. We know that $2^{\sqrt{\log_2 3}} = 2^{(\log_2 3)^{\frac{1}{\sqrt{\log_2 3}}}}$

$$= \left(2^{\log_2 3} \right)^{\frac{1}{\sqrt{\log_2 3}}} = 3^{\sqrt{\log_2 3}}$$

(Using base changing formula)

$\therefore$ The given equation becomes $x^2 - 3x + 2 = 0$

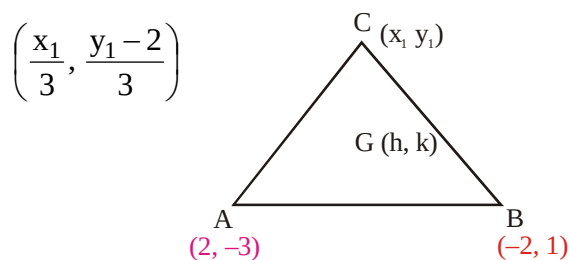
$$\Rightarrow \alpha = 1, \beta = 2$$

$$\text{Hence } \alpha^2 + \alpha\beta + \beta^2 = 7$$

4. **Ans. 9**

Sol. Let $C = (x_1, y_1)$

$$\text{The centroid is } \left(\frac{2-2+x_1}{3}, \frac{-3+1+y_1}{3} \right) \text{ i.e.}$$



It lies on the line $2x + 3y = 1$,

$$\Rightarrow \frac{2}{3}(x_1) + 3 \frac{(y_1 - 2)}{3} = 1$$

Hence the locus vertex of C is $2x + 3y = 9$

5. **Ans. 4**

Sol. The value of determinant $= 2 + 2 \sin^2 \theta$

6. **Ans. 6**

Sol. Given $b + (a_1 + a_2 + a_3 + a_4) = 8$

$$\text{or } a_1 + a_2 + a_3 + a_4 = 8 - b \quad \dots(1)$$

$$\text{Also } b^2 + a_1^2 + a_2^2 + a_3^2 + a_4^2 = 16$$

$$\text{or } a_1^2 + a_2^2 + a_3^2 + a_4^2 = 16 - b^2 \quad \dots(2)$$

Using R.M.S. $\geq$ A.M. for a_1, a_2, a_3, a_4

$$\text{or } \sqrt{\frac{a_1^2 + a_2^2 + a_3^2 + a_4^2}{4}} \geq \frac{a_1 + a_2 + a_3 + a_4}{4}$$

$$\text{or } \frac{a_1^2 + a_2^2 + a_3^2 + a_4^2}{4} \geq \frac{(a_1 + a_2 + a_3 + a_4)^2}{16}$$

$$\Rightarrow 16 - b^2 \geq \frac{(8-b)^2}{4} \quad [\text{using (1) and (2)}]$$

$$\Rightarrow 64 - 4b^2 \geq 64 + b^2 - 16b$$

$$\Rightarrow 5b^2 - 16b \leq 0$$

$$\Rightarrow b(5b - 16) \leq 0$$

$$\text{Thus } 0 \leq b \leq \frac{16}{5} \quad \text{Hence } b_{\max} = \frac{16}{5}$$

Hence possible integral values are 1, 2, 3

JEE(Main + Advanced) : NURTURE COURSE [PHASE : S, I(A), I, II, III, IV(A) & IV]**PAPER-2****PART-1 : PHYSICS****SOLUTION****SECTION-I**1. **Ans. (D)****Sol.** $\vec{v}_{0lg} = \vec{v}_1$

$$\vec{v}_{oblg} = \vec{v}_0$$

$$\vec{v}_{olob} = \vec{v}_1 - \vec{v}_0 = \vec{v}_2$$

Answer depends on angle between $\vec{v}_1$ and $\vec{v}_0$. So all are correct.

2. **Ans. (D)****Sol.** Resultant of all forces will be equal to $\frac{mv^2}{R}$ 3. **Ans. (A)****Sol.** Angular impulse = Change in angular momentum

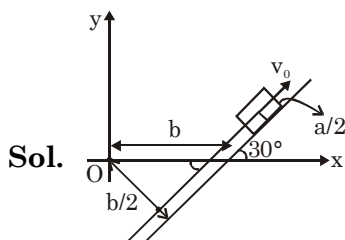
$$\frac{J\ell}{2} = \left(\frac{M}{2} \ell^2 \right) \omega$$

$$\omega = \frac{J}{m\ell} = \frac{v}{\ell}$$

4. **Ans. (D)**

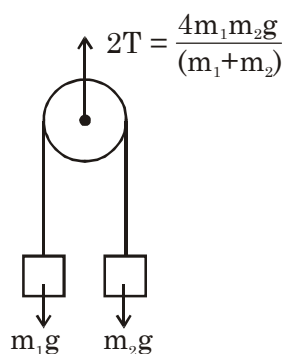
$$\text{Sol. } Mk^2 = \frac{2}{5}MR^2 - M\left(\frac{3R}{8}\right)^2 + M\left(\frac{3R}{8}\right)^2$$

$$k = \sqrt{\frac{2}{5}}R$$

5. **Ans. (D)****Sol.**

$$r = \frac{b}{2} - \frac{a}{2}$$

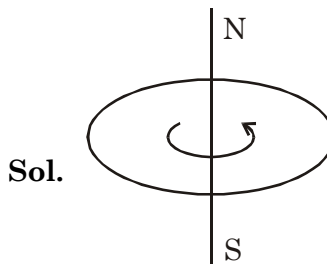
$$L = mv_0 \left(\frac{b}{2} - \frac{a}{2} \right)$$

6. **Ans. (B)****Sol.** In vertical direction gravitational force act on player-ball system.7. **Ans. (C)****Sol.**

$$\vec{a}_{cm} = \frac{\vec{F}_{ex}}{(M_{sys})}$$

$$= \frac{(m_1 + m_2)g - \frac{4m_1m_2}{(m_1 + m_2)}g}{(m_1 + m_2)}$$

$$= \left(\frac{m_1 - m_2}{m_1 + m_2} \right)^2 g \downarrow$$

8. **Ans. (C)****Sol.** $\omega = \alpha t$ α & t both same9. **Ans. (A)****Sol.**

10. Ans. (C)

Sol. At $t = 1/\sqrt{2}$ s, the body is at its highest point

$$\rho = \frac{(u \cos \theta)^2}{g} = \frac{(5\sqrt{2})^2}{10} = 5 \text{ m}$$

11. Ans. (C)

Sol. $KE_p = \frac{1}{2} m (2v)^2$

$$kE_s = \frac{3}{4} mv^2$$

12. Ans. (D)

Sol. $P = \vec{F} \cdot \vec{V}$

$$P_0 = 1000 \times 9.8 \times \frac{20}{30} \omega$$

$$P_N = 800 \times 9.8 \times \frac{30}{2} \omega$$

$$F_0 = 1000 \times 9.8$$

$$F_N = 800 \times 9.8$$

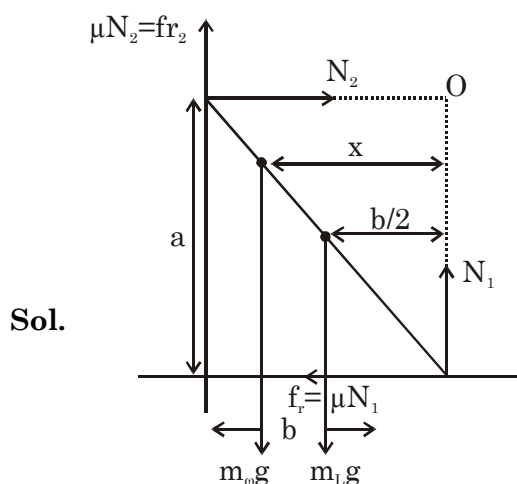
$$P_N > P_0$$

$$F_0 > F_N$$

13. Ans. (B)

Sol. $V_{cm} = \frac{6mv_0}{7m} = \frac{6v_0}{7}$, direction same.

14. Ans. (D)



Sol.

$$N + \mu N_2 = m_2 g + m_\omega g$$

$$\mu N_1 = N_2$$

$$N_1 = \frac{(m_L + m_\omega)g}{1 + \mu^2}$$

$$N_2 = \frac{\mu(m_L + m_\omega)g}{1 + \mu^2}$$

Torque about O is zero.

$$\Rightarrow \mu N_1 a + \mu N_2 b - m_\omega g x + m_L g b/2 = 0$$

$\tau_{\max}$ when $x_{\max}$ when worker on top.

15. Ans. (B)

Sol. $bx = 1$

$$b = \frac{1}{x}$$

$$= m^{-1}$$

16. Ans. (B)

Sol. Then change in internal energy may also be due to other factor such as change in the surrounding temp.

Heat dissipated = -7J

Heat gained by block < 7 J.

17. Ans. (A)

Sol. For $U_x = E$

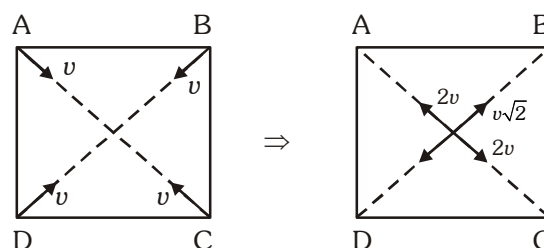
$$k = 0$$

$$v = 0$$

18. Ans. (A)

Sol. Initial momentum of sys = 0.

Final momentum can be zero only when D retraces its path with $v\sqrt{2}$.



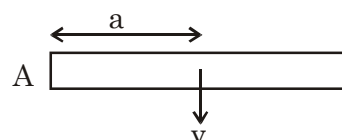
19. Ans. (A)

$$\text{Sol. } R = - \int_{r_1}^{r_2} \frac{dr}{k4\pi r^2}$$

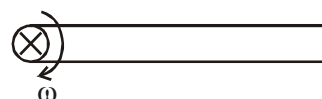
$$R \propto \left(\frac{-1}{r_1} + \frac{1}{r_2} \right)$$

$$\frac{dQ}{dt} = \frac{\Delta T}{R}$$

20. Ans. (D)



Sol.



A.M.C. about end A.

$$mva = \frac{m(2a)^2}{3} \omega$$

$$\omega = \frac{3v}{4a}$$

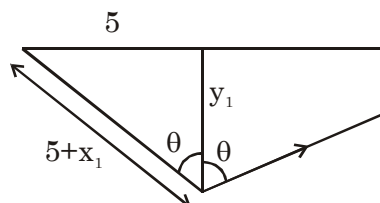
SECTION-II

1. Ans. 2.00

Sol. Binomial approximation

$$\sqrt{5^2 + y_1^2} - 5 = x_1 \Rightarrow 5 \left[1 + \frac{y_1^2}{25} \times \frac{1}{2} - 1 \right] = x_1$$

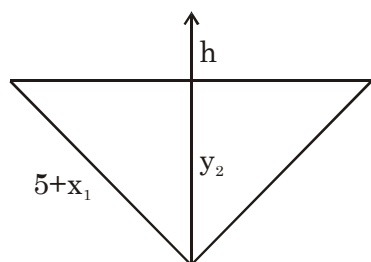
$$x_1 = \frac{y_1^2}{10}$$



$$2kx_1 \cos \theta = mg$$

$$\frac{2ky_1^2}{10} \times \frac{y_1}{\sqrt{5^2 + y_1^2}} = mg$$

$$\frac{2ky_1^3}{50 + y_1^2} = mg = \frac{2ky_1^3}{50} = \frac{ky_1^3}{25} = mg$$



$$x_2 = \frac{y_2^2}{10}$$

From W.E.T.

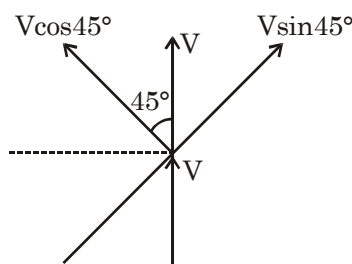
$$-\frac{2kx_2^2}{2} + mg(h + y_2) = 0$$

$$\frac{ky_2^4}{100} = \frac{ky_1^3}{25}(h + y_2)$$

$$\frac{y_2^4}{4y_1^3} - y_2 = h$$

2. Ans. 6

Sol. Just before collision



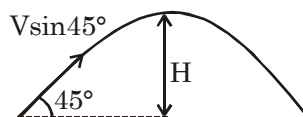
$$V^2 = (10\sqrt{3})^2 + 2(-10)(3)$$

$$= 300 - 60$$

$$V^2 = 240$$

$$V = \sqrt{240}$$

After collision



$$\Rightarrow H_{\max} = \frac{u^2 \sin^2 \theta}{2g} = \frac{(v \sin 45)^2 \sin^2 45}{2g}$$

$$= \frac{V^2 \sin^4 45^\circ}{2g}$$

$$= \frac{240}{20} \times \frac{1}{4}$$

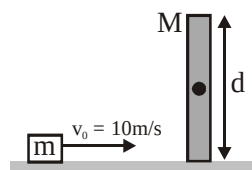
$$= 3 \text{ m}$$

$$\Rightarrow H_{\text{total}} = 3 + 3 \text{ m}$$

$$H = 6 \text{ m}$$

3. Ans. 2.00

Sol.



$$10 = \frac{d}{2} \omega_0 - v_1$$

$$m \times 10 \times \frac{d}{2} = mv_1 \frac{d}{2} + \frac{2md^2}{k} \times \omega \text{ (A.M.C.)}$$

4. Ans. 18.00

5. Ans. 5.00

Sol. Applying dimension

$$ML^2T^{-2} = M^aT^{-b}L^c$$

$$a = 1$$

$$b = 2$$

$$c = 2$$

$$\text{So } a + b + c = 5$$

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans.(C)

2. Ans.(C)

3. Ans.(A)

4. Ans.(D)

5. Ans.(C)

6. Ans.(B)

Sol. $2C(s) + O_2(g) \rightarrow 2CO(g)$

a mole —
0.2a 1.6a

$$\frac{r_{CO}}{r_{O_2}} = \frac{1.6a}{0.2a} \sqrt{\frac{32}{28}}$$

$$\frac{r_{CO}}{r_{O_2}} = 8\sqrt{\frac{8}{7}} = \sqrt{\frac{8^3}{7}}$$

7. Ans.(A)

8. Ans.(C)

9. Ans.(C)

10. Ans.(C)

11. Ans.(D)

12. Ans.(D)

$$u_{rms} = 1.22 \times u_{np} = 488 \text{ m/s}$$

$$u_{Avg} = 1.13 \times u_{mp} = 452 \text{ m/s}$$

13. Ans.(C)

14. Ans. (B)

15. Ans.(C)

$$E(\text{photon}) = \frac{1240}{155} = 8\text{eV}$$

$$w = \frac{64 \times 10^{-20}}{1.6 \times 10^{-14}} = 4\text{eV}$$

$$\text{K.E. of electron} = 8 - 4 = 4\text{eV}$$

$$\lambda = \frac{1240}{4} = 310\text{nm}$$

16. Ans.(C)

17. Ans. (D)

Compounds should have same MF to show isomerism.

$C-C-C-C$ & $C-\overset{\overset{C}{|}}{\underset{\underset{C}{|}}{C}}-C$ are isomer (chain isomer)

18. Ans. (D)

2nd line balmer series ($4 \rightarrow 2$) H-atom

$$\frac{1}{\lambda} = R.1^2 \left(\frac{1}{2^2} - \frac{1}{4^2} \right)$$

For Li^{2+}

$$\frac{1}{\lambda} = R.3^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$\frac{1}{\lambda} = R.1^2 \left(\frac{1}{(n_1/3)^2} - \frac{1}{(n_2/3)^2} \right)$$

$$\frac{n_1}{3} = 2 ; \frac{n_2}{3} = 4$$

$$n_1 = 6 ; n_2 = 12$$

19. Ans.(D)

20. Ans.(D)

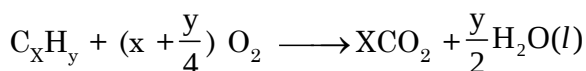
SECTION-II

1. Ans.(1.25)

2. Ans.(9.00)

3. Ans.(9.00)

4. Ans.(12.00)



$$\text{Vol } a \quad a \left(x + \frac{y}{4} \right) \quad ax$$

Given that volume of $O_2 = 6$ vol of C_xH_y

$$a(x + y/4) = 6a$$

$$x + \frac{y}{4} = 6 \quad \dots (1)$$

vol of $CO_2 = 4$ vol of C_xH_y

$$ax = 4 \text{ (a)}$$

$$x = 4 \quad \dots (2)$$

$$\text{from (1)} \quad x + \frac{y}{4} = 6$$

$$4 + \frac{y}{4} = 6$$

$$y = 8$$

$$\therefore x + y = 4 + 8 = 12$$

5. Ans.(2.00)

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (A)**

Equation of AB is $y - 1 = m(x - 1)$

$\Rightarrow$ Equation of AC is $y - 1 = -\frac{1}{m}(x - 1)$

$$B\left(1 - \frac{1}{m}, 0\right)$$

$$C\left(0, 1 + \frac{1}{m}\right)$$

D is mid point of BC

$$\Rightarrow 2h = 1 - \frac{1}{m}$$

$$\text{and } 2k = 1 + \frac{1}{m}$$

Eliminating 'm' we get locus of Δ is $x + y = 1$

2. **Ans. (D)**

We have $\sin^3 10^\circ + \sin^3 50^\circ - \sin^3 70^\circ$

$$= \frac{1}{4}[(3\sin 10^\circ - \sin 30^\circ) + (3\sin 50^\circ - \sin 150^\circ) - (3\sin 70^\circ - \sin 120^\circ)]$$

$$= \frac{1}{4}\left[3(\sin 10^\circ + \sin 50^\circ - \sin 70^\circ) - \frac{3}{2}\right]$$

$$= \frac{1}{4}\left[3(\sin 10^\circ - 2\cos 60^\circ \cdot \sin 10^\circ) - \frac{3}{2}\right] = -\frac{3}{8}$$

3. **Ans. (D)**

Focus of parabola $y^2 = 4ax$ is $(a, 0)$ the line $y - x - a = 0$ pass through $(a, 0)$ so it is a focal chord. Hence the tangents intersects at 90° .

4. **Ans. (A)**

$$\text{Given } \cot \frac{A}{2} \cdot \cot \frac{C}{2} = 3$$

$$\Rightarrow \frac{\cos \frac{A}{2} \cdot \cos \frac{C}{2}}{\sin \frac{A}{2} \cdot \sin \frac{C}{2}} = 3 \Rightarrow \frac{\cos \frac{A+C}{2}}{\cos \frac{A-C}{2}} = 2$$

(using componendo and dividendo)

$$\Rightarrow \frac{2\sin \frac{A+C}{2} \cos \frac{A-C}{2}}{2\sin \frac{A+C}{2} \cdot \cos \frac{A+C}{2}} = 2$$

$$= 2\sin B = \sin A + \sin C$$

5. **Ans. (C)**

Equation of tangent to ellipse at

$$P(5\cos\theta, 4\sin\theta) \text{ is } \frac{x}{5}\cos\theta + \frac{y}{4}\sin\theta = 1, \text{ it}$$

meets $x = 0$ at $Q \Rightarrow Q$ is $(0, 4\operatorname{cosec}\theta)$, its image in $y = x$ is $R(4\operatorname{cosec}\theta, 0)$ so equation of circle with QR as diameter is

$$x(x - 4\operatorname{cosec}\theta) + y(y - \operatorname{cosec}\theta) = 0$$

$$\Rightarrow x^2 + y^2 - 4(x + y)\operatorname{cosec}\theta = 0$$

$\Rightarrow$ each member of the family pass through the intersection of $x^2 + y^2 = 0$ and $x + y = 0$, so the point is $(0, 0)$

6. **Ans. (A)**

$$\cos^2 \alpha + \cos^2(\alpha + 120^\circ) + \cos^2(\alpha - 120^\circ)$$

$$= \cos^2 \alpha + \left\{\cos(\alpha + 120^\circ) + \cos(\alpha - 120^\circ)\right\}^2 - 2\cos(\alpha + 120^\circ)\cos(\alpha - 120^\circ)$$

$$= \cos^2 \alpha + \left\{2\cos\alpha\cos 120^\circ\right\}^2 - 2\{\cos^2 \alpha - \sin^2 120^\circ\}$$

$$= \cos^2 \alpha + \cos^2 \alpha - 2\cos^2 \alpha + 2\sin^2 120^\circ$$

$$= 2\sin^2 120^\circ = 2 \times \frac{3}{4} = \frac{3}{2}$$

7. **Ans. (A)**

Common chord of two intersecting circles is $\perp^r$ to line of centres

8. **Ans. (C)**

Sol. The chord forms an equilateral Δ with one vertex at the centre of circle so that its midpoint is at a constant distance of

$$\frac{\sqrt{3}}{2} \text{ times the radius from the centre.}$$

The locus is

$$(x^2 - 1)^2 + (y - 2)^2 = \frac{3}{4} (16)$$

9. **Ans. (B)**

Equation tangent at $(0, 1)$ to the circle $x^2 + y^2 = 1$

$$y = 1$$

equation line passing through $(3, 1)$ is

$y - 1 = m(x - 3)$. if is. also the tangent to the circle.

$$p = r$$

$$\Rightarrow \left| \frac{1 - 3m}{\sqrt{1 + m^2}} \right| = 1$$

$$\Rightarrow m = \frac{3}{4} \text{ so. equation } 3x - 4y - 5 = 0$$

10. Ans. (A)

We have $q = -p - \frac{2}{p} \Rightarrow p^2 + pq + 2 = 0$

11. Ans. (B)

AB is common chord of $x^2 + y^2 = 13$ and circle having centre at p and radius $\sqrt{5}$.

$\Rightarrow (x-3)^2 + (y-2)^2 = 5 \quad \dots(1)$

equation of common chord

$\Rightarrow S_1 - S_2 = 0$

$\Rightarrow 6x + 4y - 21 = 0$

12. Ans. (A)

$S = 19 \times \left(\frac{2+102}{2} \right) = 19 \times 52$

$d = \frac{2-102}{19+1} = -5$

$A_{17} = a + 17d = 102 + 17(-5) = 17$

$A_{10} = a + 10d = 102 + 10(-5) = 52$

$\therefore \frac{S}{(A_{17}+2)A_{10}} = \frac{19 \times 52}{(17+2) \times 52} = 1$

13. Ans. (C)

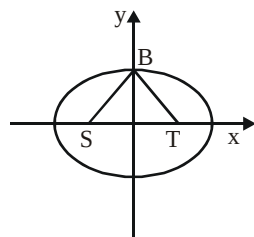
$\frac{1}{\sqrt{3}} = \frac{2at}{at^2}$

$t = 2\sqrt{3}$

Length of side of equilateral

$\Delta = 4at = 4(4)(2\sqrt{3}) = 32\sqrt{3}$

14. Ans. (C)



Let the equation of ellipse be $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

$S \equiv (-ae, 0)$

$T \equiv (ae, 0)$

$B \equiv (0, b) \Rightarrow SB = \sqrt{(0+ae)^2 + b^2}$

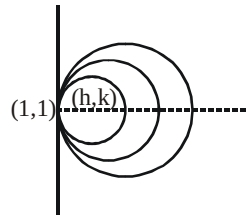
Also $SB^2 = ST^2 \Rightarrow 4a^2e^2 = a^2e^2 + b^2$

or $3a^2e^2 = a^2(1-e^2)$

$= a^2 - a^2e^2 \Rightarrow e^2 = 1/4 \Rightarrow e = 1/2$

Hence (C) is the correct answer.

15. Ans. (B)



Locus of the center of the circle is a line perpendicular to the given line

$\Rightarrow \frac{k-1}{h-1} = -\frac{1}{2} \Rightarrow x+2y=0$

16. Ans. (C)

Put $x-1=t^2$ in the given equation, we get

$\sqrt{t^2+1+3-4t} + \sqrt{t^2+1+8-6t} = 1$

$\Rightarrow |t-2| + |t-3| = 1$

$\Rightarrow t \in [2, 3]$

$\Rightarrow x \in [5, 10]$

17. Ans. (B)

$\sin^4\theta + \cos^4\theta = 1 - 2\sin^2\theta\cos^2\theta$

$= 1 - \frac{1}{2}\sin^2 2\theta$

$\Rightarrow \text{max value} = 1$

18. Ans. (B)

$\sin\alpha, \sin\beta, \cos\alpha$ are in G.P.

$\Rightarrow \sin^2\beta = \sin\alpha \cos\alpha$

$\Rightarrow \cos 2\beta = 1 - \sin 2\beta \geq 0$

Now, the discriminant of the given equation is

$4\cot^2\beta - 4 = 4\cos 2\beta \cdot \text{cosec}^2\beta \geq 0 \Rightarrow \text{Roots are always real.}$

Hence (B) is the correct answer.

19. Ans. (D)

$\sin x \cos x [\sin^2 x + \sin x \cos x + \cos^2 x] = 1$

$\Rightarrow \sin x \cos x + (\sin x \cos x)^2 = 1$

$\sin^2 2x + 2\sin 2x - 4 = 0$

$\Rightarrow \sin 2x = \frac{-2 \pm \sqrt{4+16}}{2} = -1 \pm \sqrt{5}$

which is not possible.

Hence (D) is the correct answer.

20. Ans. (D)

$$\begin{vmatrix} x^2 + y^2 & ax + by & px + qy \\ ax + by & a^2 + b^2 & ap + bq \\ px + qy & ap + bq & p^2 + q^2 \end{vmatrix}$$

$$= \begin{vmatrix} x & y & 0 \\ a & b & 0 \\ p & q & 0 \end{vmatrix} \times \begin{vmatrix} x & a & p \\ y & b & q \\ 0 & 0 & 0 \end{vmatrix} = 0$$

SECTION-II

1. Ans. 1.00

$$\text{Given } 2 \sin x + 2 \sin 2x - 1 = 1$$

$$\text{Or, } \sin^2 x + \sin x - 1 = 0$$

$$\therefore \sin x = \frac{-1 \pm \sqrt{1+4}}{2} = \frac{-1 + \sqrt{5}}{2}$$

$$\Rightarrow \sin^2 x = \frac{3 - \sqrt{5}}{2} \Rightarrow \cos^2 x = \frac{\sqrt{5} - 1}{2}$$

$$\therefore \cos^2 x (1 + \cos^2 x) = \frac{\sqrt{5} - 1}{2} \times \frac{\sqrt{5} + 1}{2} = 1$$

2. Ans. 3.00

$$\frac{\cos 16 \cos 44}{\sin 16 \sin 44} - 1 + \frac{\cos 44 \cos 76}{\sin 44 \sin 76} - 1 - \frac{\cos 76 \cos 16}{\sin 76 \sin 16} - 1 + 3$$

$$= \frac{\cos 60}{\sin 16 \sin 44} + \frac{\cos 120}{\sin 44 \sin 76} - \frac{\cos 60}{\sin 76 \sin 16} + 3$$

$$= \frac{1}{2} \left(\frac{\sin 76 - \sin 16}{\sin 16 \sin 44 \sin 76} \right) - \frac{1}{2 \sin 76 \sin 16} + 3 = 3.$$

3. Ans. 4.00

The tangent of slope m must be of the form

$$y = m(x + 2) + \frac{a}{m}$$

$$\text{So, } 2m + \frac{2}{m} = c \Rightarrow c = 2$$

$$c = 2 \left(m + \frac{1}{m} \right) \geq 2 \times 2. \text{ So } c_{\min} = 4$$

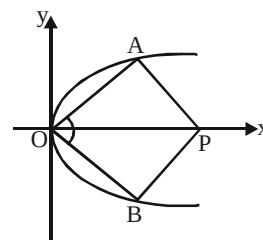
4. Ans. 24.00

$$\text{Let } A \equiv (t^2, 2t), B \equiv (t^2, -2t)$$

$$m_{OA} = \frac{2}{t}, m_{OB} = -\frac{2}{t}$$

$$\Rightarrow t^2 = 4$$

$$\Rightarrow t = 2$$



Equation of normal AP is

$$y = -2x + 4 + 8$$

$$\Rightarrow P \equiv (6, 0)$$

Thus area of quadrilateral

$$OAPB = \frac{1}{2} (OP)(AB)$$

$$= \frac{1}{2} \cdot 6 \cdot 8 = 24 \text{ sq. units.}$$

5. Ans. 0.00

$$\begin{vmatrix} 1 & 1 & -1 \\ m-1 & m^2-7 & -5 \\ m-2 & 2m-5 & 0 \end{vmatrix} = 0$$

$$m^3 - 4m^2 + 5m - 6 = 0$$

$$(m^2 - m + 2)(m - 3) = 0$$

$$D < 0 \text{ of } m^2 - m + 2 = 0$$

so, $m = 3$, but

at $m = 3$ lines are parallel, so no. of values of m is zero.